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Original article

The deep learning method applied to the detection and mapping of stone deterioration in open-air sanctuaries of the Hittite period in Anatolia ☆

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ABSTRACT

The detection of deterioration in archeological heritage sites is a very time-consuming task that requires expertise. Furthermore, vision-based approaches can cause errors, considering the complex types of deterioration that develop in different scales and forms in monuments. This problem can be solved effectively owing to computer vision algorithms, commonly used in different areas nowadays. This study aims to develop a model that automatically detects and maps deteriorations (biological colonization, contour scaling, crack, higher plant, impact damage, microkarst, missing part) and restoration interventions using the Mask R-CNN algorithm, which has recently come to the fore with its feature of recognizing small and large-sized objects. To this end, a total of 2460 images of Yazılıkaya monuments in the Hattusa archeological site, which is on the UNESCO heritage list, were gathered. In the training phase of the proposed method, it was trained in model 1 to distinguish deposit deterioration commonly observed on the surface of monuments from other anomalies. Other anomalies trained were model 2. In this phase of the models, the average precision values with high accuracy rates ranging from 89.624% to 100% were obtained for the deterioration classes. The developed algorithms were tested on 4 different rock reliefs in Yazılıkaya, which were not used in the training phase. In addition, an image of the Eflatunpınar water monument, which is on the UNESCO tentative list, was used to test the model's universality. According to the test results, it was determined that the models could be successfully applied to obtain maps of deterioration and restoration interventions in monuments in different regions.

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1. Introduction

Archeological sites are heritage areas that enlighten the collective memories of past societies, such as their habitats, esthetic concerns, and beliefs. On movable and immovable cultural heritage, the conservation of stone, which is the most used material in these areas helping reconstruct history, is threatened by various deterioration processes due to natural or anthropogenic causes. When movable cultural heritage (such as statues, tablets and jewelry, cookware) is brought to the surface by archeological excavations, the process of identifying problems in works and planning restoration interventions can be quite fast. In immovable cultural heritage (palaces, churches, rock reliefs, etc.), periodic inspections should be performed in addition to this process. If the necessary inspection and measures are not taken in both types of heritage, a small-scale (e.g. flaking, spalling, cracking) deterioration type can turn into a larger-scale anomaly (such as contour scaling, differential erosion, missing part) over time, and this may cause the loss of an archeological trace. Therefore, the identification of deterioration in the preservation of heritage in archeological sites constitutes one of the most important stages of restoration studies. However, the size of these areas makes it difficult to examine the types of deterioration in each work in periodic time frames. Moreover, the current inspections performed by experts by on-site visual analysis can also lead to human-induced errors. Developing computer-aided methods may be an important option in solving this situation. To reduce the workforce and margin of error in archeological sites and achieve long-term comparability on-site as well as among

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different sites, computer-aided methods such as remote sensing [1], relief extraction [2], automatic drawing [3], and photogrammetry [4] were first used. Afterward, by supporting devices such as terrestrial laser scanning and unmanned aerial vehicles with computer software, digital reconstruction [5], surface erosion examination [6], and pixel-based archeological feature detection [7] studies have been conducted. However, although all these methods have significant advantages in heritage sites, they have inadequacies in identifying deterioration. The most important one of these inadequacies is that deterioration groups are different from each other in shape, texture, and color. Therefore, expert opinion is needed to interpret the data during the identification. Other disadvantages of these methods are that they are expensive and their data processing time is long. To overcome these problems, the deep learning (DL) method, which can learn human skills with a large number of data entries, can be an effective tool. It is possible to create object recognition algorithms with high accuracy and precision with DL, an artificial intelligence method. With this feature, some studies [8–10] have been recently conducted in archeological sites using DL, which has become widespread in different disciplines. In their study, Bewes et al. [8] predicted sex from skeletal remains using DL and obtained 95% accuracy values. Verschoof-van der Vaart and Lambers [9] identified prehistoric barrows and Celtic fields using the Faster R-CNN algorithm. Cintas et al. [10] used the pottery profile classification method and obtained a mean accuracy score of 0.9013. Apart from these studies, there are also damage assessment studies in the limited components of historic monuments (brick walls, building stones, and tiles) [11–14]. In their study, Wang et al. [11] tried to determine spalling and efflorescence type deterioration in a historic brick wall in China using the Faster R-CNN algorithm. Hatır et al. [12] classified eight different deterioration types observed in building stones in the Konya region and fresh rock using the DL method in their study. Wang et al. [13] determined the damage observed in historic glazed tiles using the Mask R-CNN algorithm. In their study, Hatır and İnce [14] demonstrated that the lithologies of the building stones used on the southern facade of the Matbah-ı Şerif monument could be determined by the DL method. These studies in the literature show that artificial intelligence algorithms yield extremely important results in the fields of anthropology, archeology, and structural engineering.

In previous studies, applications in which a limited amount of deterioration (spalling, efflorescence, and damage) was detected in building materials were made. However, there are no studies on reliefs where detection is more difficult. In addition, there are deficiencies in the literature for determining the types of deterioration in archeological sites via computer vision. To eliminate these deficiencies, automatic deterioration detection and mapping using the Mask R-CNN method in reliefs, an integral part of cultural heritage, is targeted.

2. Research aim

This study aims to develop a DL algorithm that can generate the automatic mapping of the types of deterioration in archeological heritage sites with fast and accurate detection. To this end, seven different types of deterioration (missing part, contour scaling, crack, microkarst, impact damage, biological colonization, higher plant), which are generally observed in the open-air sanctuary of Yazılıkaya located nearby the Hattusa (Çorum, Turkey) archeological site and inscribed on the UNESCO cultural heritage list, and restoration interventions were primarily trained in the Mask R-CNN algorithm. Then, this algorithm was tested on four unique Hittite Yazılıkaya reliefs not used in training, and automatic deterioration maps were obtained. In addition, to evaluate the success of the developed algorithm in different environmental factors and lithologies, the image of the Eflatunpınar Hittite Water monument (Konya, Turkey) was used. This monument was included in the UNESCO's World Heritage Tentative List in 2014. The results obtained from the study indicate that the proposed algorithm can produce deterioration maps in archeological sites with high accuracy and precision.

3. Archeological frameworks of the study

The archeological site of Boğazköy has been identified as the ancient city of Hattusa, the capital city of the Hittite Empire, one of the major political powers during the 2nd millennium BC in the Near East by excavations and research ongoing since 1906 [15]. The unearthed monumental structures within the city reveal major insights into the art and history of the first state in Anatolia. The open-air rock sanctuary of Yazılıkaya is located approximately 2 km northeast of the city (Fig. 1). Although archeological findings indicate that the rocky outcrops were used for religious purposes already in the 3rd millennium BC, the present state of the monument was executed under the rule of the Great King Tuthaliya IV in the mid-13th century BC [16]. The sanctuary consists of two major spaces (chamber A and B), which are decorated with more than 90 depictions of the Hittite gods carved out of the rocks (Fig. 1a). In chamber A, a gathering of the gods of the Hittite pantheon is shown; the male gods are strung one behind the other on the left side of the entering visitor (Fig. 1e, k-l), whereas the female goddesses are depicted on the right (Fig. 1d, m). Each god is identified by typical attributes and an inscription in Luwian hieroglyphs indicating his/her name. At the front side, both rows meet in the largest relief showing a couple of the highest gods (Fig. 1g): the male storm-god Tešub (left) and his female counterpart Hepat (right) [16]. Chamber B instead is decorated with scenes related most probably to the transition of the Hittite king after his death (Fig. 1c, i-h). The Great King Tuthaliya IV himself is shown in a larger-than-life relief in chamber A (Fig. 1j). Together with the building remains in front of the rock chambers, the site represents the typical features of a Hittite temple as excavated in Hattusa and elsewhere (Fig. 1a) [16]. The functions of both chambers are still much debated. Chamber A may have played a role in the yearly spring festivals marking the beginning of the new year and possibly during the enthronement of a new king. While chamber A shows slight traces of earlier decorations and thus indicates use during the whole Hittite period (c. 1650/00–1180 BCE), chamber B seems to have been exclusively commissioned by the Great King Tuthaliya IV (Fig. 1c, i), whose transition to the netherworld might be shown here by the representation of the god Nergal (Fig. 1h).

Hattusa and Yazılıkaya were inscribed in the UNESCO World Heritage List in 1986 under the title of "Hattusa: the Hittite Capital" with its different types of architectural monuments and rich ornamental features such as the Lions' Gate, Kingś Gate, and the reliefs carved into the bedrock of Yazılıkaya.

Furthermore, one of the unique structures belonging to the Hittite Empire, which was extremely effective in the cultural development of Anatolia, is the Eflatunpınar water monument. This monument, located approximately 90 km west of Konya province, is one of the works of the last period of the Hittite Empire. The monument, in which figures of gods, goddesses, and hybrid creatures are carved on monolithic stones, is 5.20 m high and 6.62 m wide. Water flows into the 30 × 35 m holy pool from the 6 holes in the figures below the monument [17].

3.1. Geological setting of the monuments

Hattusa is geologically located close to the Izmir-Ankara-Erzincan Suture Zone, and the city was founded on the Ankara

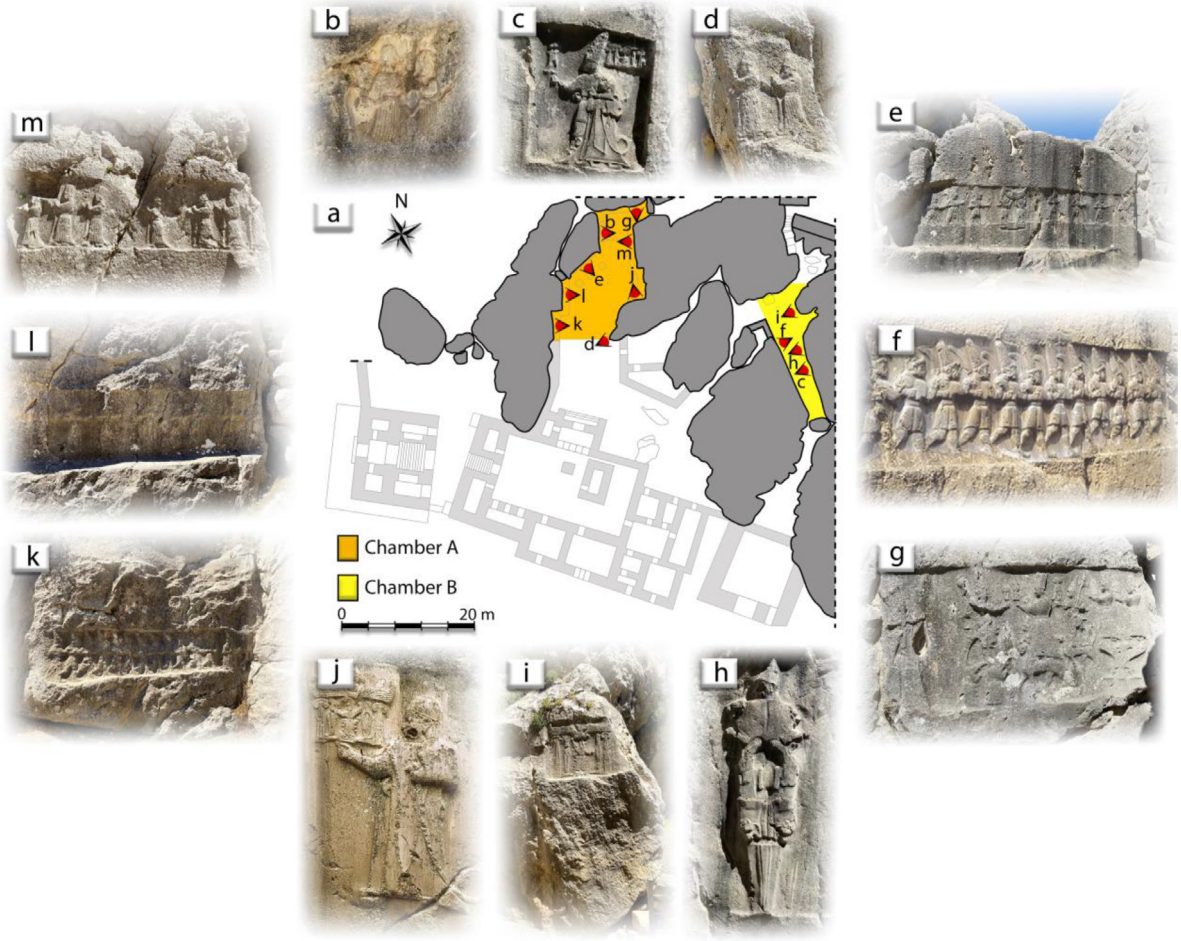


Fig. 1. The plan of the rock sancturay of Yazılıkaya and samples of the reliefs, (a) Yazılıkaya site plan, (b) two femal goddesses following the goddess Ishtar in Chamber A, (c) god Šarumma and king Tuthaliya IV., (d) two femal goddesses, (e) a part of the male gods in chamer A, (f) the group of 12 running gods in chamber B, (g) the main scene, showing the meeting of the major female (left) and male (right) gods of the Hittite pantheon, (h) the so-called sword god, representing the Mesopotamian god of the netherworld Nergal, (i) relief of the name of the great king Tuthaliya IV written in Luwian Hieroglyphs, (j) the relief of great king Tuthaliya IV., (k) the group of 12 running gods at the beginning of the row of male gods in chamber A, (l) the row of male gods in chamer A, (m) the row of the femal goddesses in chamber A.

Melange. These lithodemic units, which expose a large area between the Ankara and Yozgat regions, were mapped by Blumenthal [18] as mixed Mesozoic tectonic facies. These rock assemblages were first named by Bailey and McCallien [19] as the Ankara Melange. Ketin [20] described the units as Upper Cretaceous formations and indicated that blocks of Palaeozoic and Lower Mesozoic age were contained within the melange. Norman [21] distinguished several sub-belts and lenses within the Ankara Melange Zone and divided the rock units into six main groups: the melange with metamorphic, ophiolitic and limestone blocks, ocean floor fragments, flysch and acidic/intermediate magmatic rocks. The best crops of the unique olistostromal section of the Ankara Melange, which is spreading from Ankara to Sivas in a very wide area, are located especially around Hattusa. Kazancı et al. [22] suggested this location as a geological heritage with the name of Hattusa Olistoliths. The rock units in the study area are geologically divided into two main groups as the Ankara Melange and Miocene-Pliocene continental rocks (Fig. 2). The Ankara Melange, where the main city was established, consists of three main rock units. These are a matrix of the melange, blocks and olistoliths of limestone and a tectonic slice of serpentines. The melange matrix is predominantly composed of fine-graded clastic sedimentary rocks and pelagic carbonates, which were mostly deposited in flysch facies of the Neotethyan Ocean. The melange matrix also consists of 100-meter-high olistolithic recrystallized limestones and mesoscopic-scaled metaserpentinite and metabasites as block or tectonic slices (Fig. 2). The Yazlıkaya reliefs are located on these limestone olistoliths. Limestones contain rich biogenic components, according to Dahm [23], and are named biomicrite. The use of these olistoliths as quarries for the city's construction minimizes the efforts to move building blocks from the outside. The combination of these geological features is among the reasons to support Hattusa's location at this point [24].

There are geological units around the Eflatunpınar water monument, one of the Hittite works of art, from the Cambrian to Quaternary (Fig. 2) [25]. The basis of the region consists of Paleozoic-Mesozoic sediments composed of sandstone, conglomerate, dolomite, limestone, and dolomitic limestone. The Upper Miocene-aged Kızılören formation, consisting of conglomerate, sandstone, mudstone, tuffite, tuff, ignimbrite, and volcanic breccia, unconformably overlies these units. The Dilekçi Formation, consisting of the Upper Miocene-Pliocene-aged volcano-sedimentary sequence, comes over the Kızören Formation with a lateral vertical transition. The lower units are cut by the Plio-Quaternary-aged, gray-dark gray-colored, fine to medium crystalline Selki andesite [25]. All units in the study area are unconformably covered by Quaternary-aged alluvium and slope debris (Fig. 2). Bozdag et al. [4] stated that the building blocks of the monument were of dacite composition.

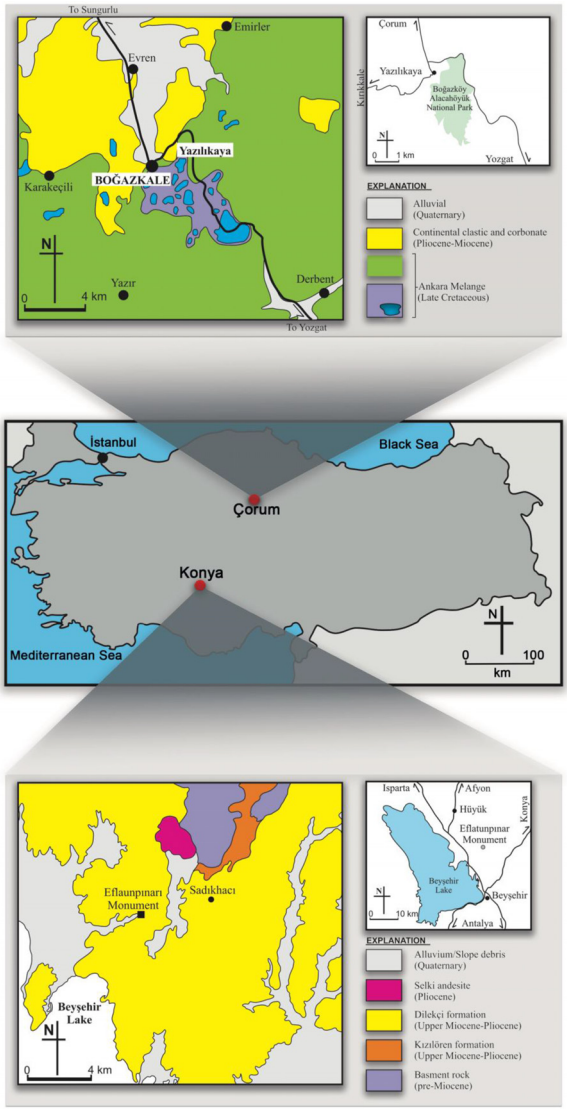


Fig. 2. Location and simplified geological map of the study areas [22,25].

3.2. Characterizing the deterioration of the monuments

The types of deterioration developing in stones used in historic monuments as a result of atmospheric, anthropogenic, and biological processes have been classified by many researchers [26–28]. In this study, the classification of the deterioration types observed in the Yazılkaya rock and Eflatunpınar water monuments were made according to International Council on Monuments and Sites-International Scientific Committee for Stone (ICOMOS-ISCS) [28]. According to Schachner [24], the presence of coloring agents such as pigment application was not found in the Yazılıkaya stone reliefs. In this study, taking the mentioned data into account, the types of deterioration and restoration practices in the region are briefly described below.

- Crack: This type of deterioration occurs as a result of atmospheric processes (wetting-drying, salt crystallization, freeze-thaw, and thermal shock) and mechanical events (static conditions such as load reduction and increase or earthquake). This process accelerates crack development by increasing the water content in the rock body [12,29] (Fig. 3a).
- Contour scaling: It is the separation of the upper part of the rock, which is deteriorated as a result of atmospheric effects, into layers with thicknesses varying from mm to cm in parallel to the solid substract part [29] (Fig. 3b).
- Missing part: It is the loss of parts occurring as a result of weaknesses in engineering properties in a certain part of the stone due to atmospheric effects [28] (Fig. 3c).
- Impact damage: It is crater-shaped material losses caused by anthropogenic effects on the stone surface [28] (Fig. 3d).
- Microkarst: It is concavities on scales varying from mm to cm, revealed by the dissolution process on carbonate rock surfaces [28] (Fig. 3e).
- Rounding: It is formed when the stone loses its angular edge form due to atmospheric effects [28] (Fig. 3f).
- Biological colonization: It is the development of biological formations such as fungi, lichen, and algae under appropriate conditions (humidity and temperature) due to the accumulation of organic matter on these surfaces as a result of increasing roughness with the deterioration of the rock surface over time [30] (Fig. 3g).
- Higher plant: It is the development of rooted plants with the increasing amount of organic matter accumulated in the gaps and cracks developing in rocks over time. In this process, as plants grow, the pressure created by their roots increases and creates the mechanical fragmentation process [30] (Fig. 3h).
- Deposit: It is the accumulation of exogenic materials such as atmospheric dust, sea salt aerosols, and lime layers on the stone surface [28] (Fig. 3i).
- Restoration intervention: These are practices such as consolidation, patch treatment, or integration to prevent the development of deterioration types on the monument surface. A filling of cracks was performed at Yazılıkaya to consolidate the rocks (Fig. 3j).

4. Material and method

Artificial intelligence algorithms, which basically try to imitate human skills, have recently shown significant developments. Deep learning, one of these algorithms, is a method inspired by the process that a person sees and processes in his brain. The study performed by LeCun et al. [31] and based on the principle of moving pixels on the image via Convolutional Neural Networks (CNNs) constitutes the first step of developments in this method. However, this algorithm can perform only rough localization in object recognition [32]. To overcome this shortcoming, Girshick et al. [33] developed regions with the convolutional neural networks (R-CNN) algorithm. However, the long data evaluation process (training and testing) of this method is its biggest disadvantage. The Fast R-CNN algorithm, in which the image is directly transferred to CNN, was proposed by Girshick [32] to solve this problem. In the recent study, the Faster R-CNN method was developed, enabling the further reduction of testing and training times through the Region Proposal Network (RPN) [34]. Object detection can be performed with high accuracy and precision using the Faster R-CNN algorithm, one of the most frequently used methods in the literature. However, since object detection is carried out with the bounding box in this method, quite complex results can be obtained in cases with a high number of objects. To overcome this, the Mask R-CNN algorithm has been developed owing to the extra branch added to the Faster R-CNN algorithm. It is possible to perform instance segmentation studies owing to the mask applied by Mask R-CNN for each object suggestion [35].

Mask R-CNN is a method that integrates CNNs into region proposals, performs tests with end-to-end training and applies masks to detected objects. In this way, the success of training and testing increase exponentially, and real perception is imitated. The structure of the Mask R-CNN method used in this study is presented in Fig. 4. Upon Fig. 4, the input image is first transferred

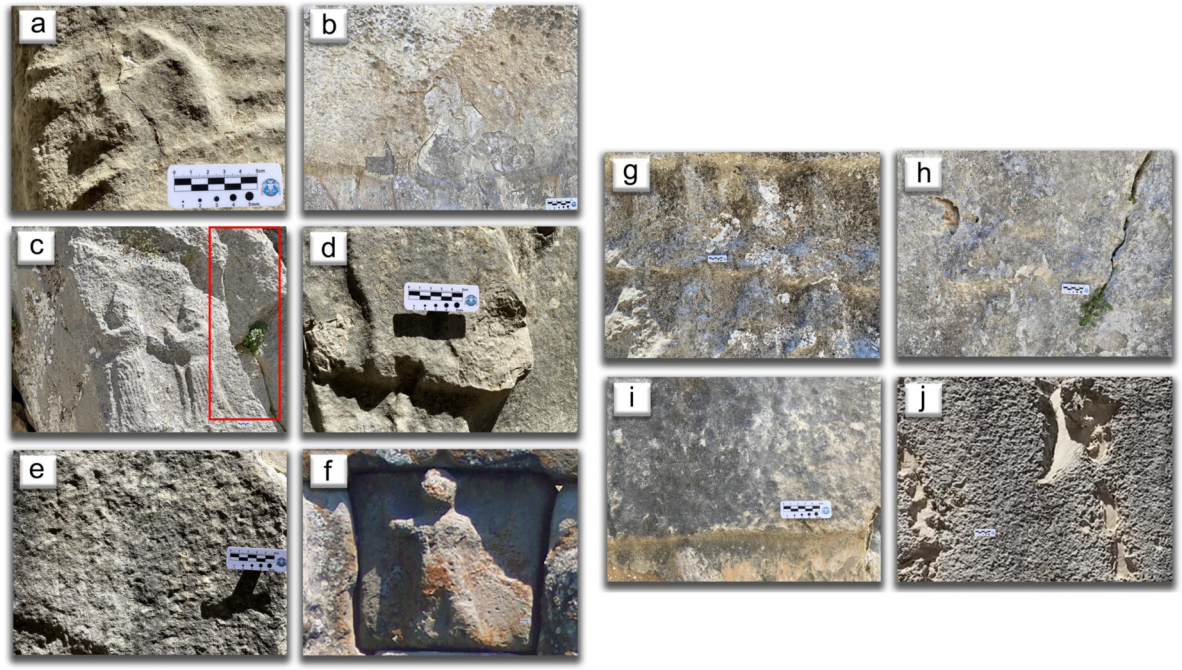


Fig. 3. Deterioration features and restoration practice in the Hattuşa archeological site; (a) crack, (b) contour scaling, (c) missing part, (d) impact damage, (e) microkarst, (f) biological colonization, (g) higher plant, (h) deposit, (i) restoration intervention.

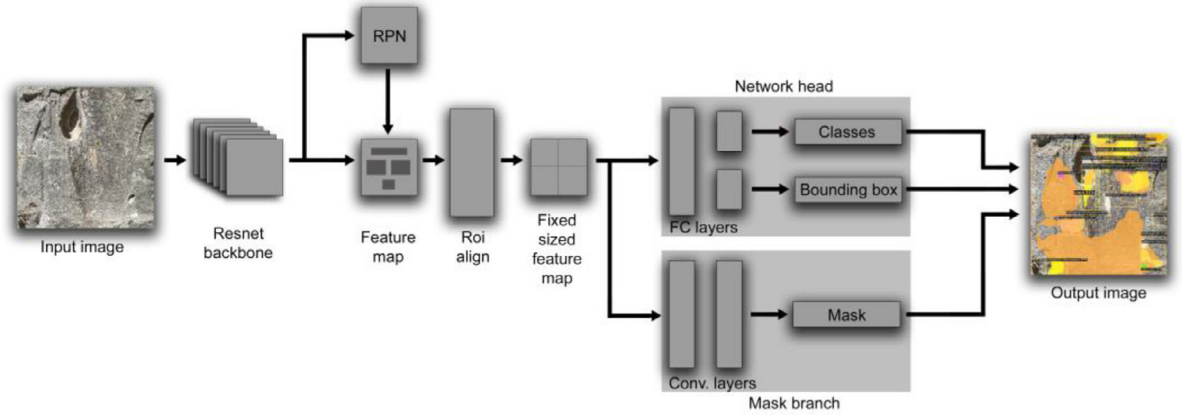


Fig. 4. The structure of Mask R-CNN Algorithm.

to the backbone network. In the first layers of the backbone network, lower-level features (such as edges and corners) are perceived, while more complex features (such as people, cars, and deterioration anomalies) are perceived in the advanced layers. In this network, the ResNet-101 model was used, which allowed for reducing the training time of the model. Furthermore, the feature was strengthened with the pyramid network (FPN) to increase the success of this network in detecting small objects. Afterward, the feature maps obtained from this network were sent to RPN to receive region offers and reduce the time in determining the recommended window size. RPN is an anchor approach that creates rectangular boxes on the image using specific aspect ratios. In this approach, each point in the feature map matches in the original image with nine links (Fig. 5). With pixel dimensions of 6×6 in Fig. 5, the number of anchors to create the feature map is $6 \times 6 \times 9 = 324$. With this feature in RPN, object ratio differences in images at different distances and angles are overcome. The obtained feature maps are transferred to ROI (region of interest) align so that they can be brought to fixed dimensions. In the next step, the object coordinates and categories in the feature maps are determined by bbox regression and softmax, respectively, in the FC layer. Finally, the object is segmented and transferred to the Mask branch for the mask application function.

4.1. Implementation details

The images of all training data in this study were compiled from the Hattusa Yazılıkaya area. The images were obtained by a 12-megapixel (MP) resolution digital camera in 4000 × 3000 dimensions and 72dpi in the RGB format. Since the deposit, one of the types of deterioration commonly observed in the study area, had large surface areas, it was thought to be likely to cover the results of other classes in the testing stage. To overcome this situation and increase the visuality of the results, the deposit anomaly was evaluated as one class (model 1), and other types of deterioration and restoration intervention were evaluated as another class (model 2).

To create the model, data sets belonging to the training, validation, and test stages are needed. Training is the stage of making inferences by processing the existing attributes in the labeled im-

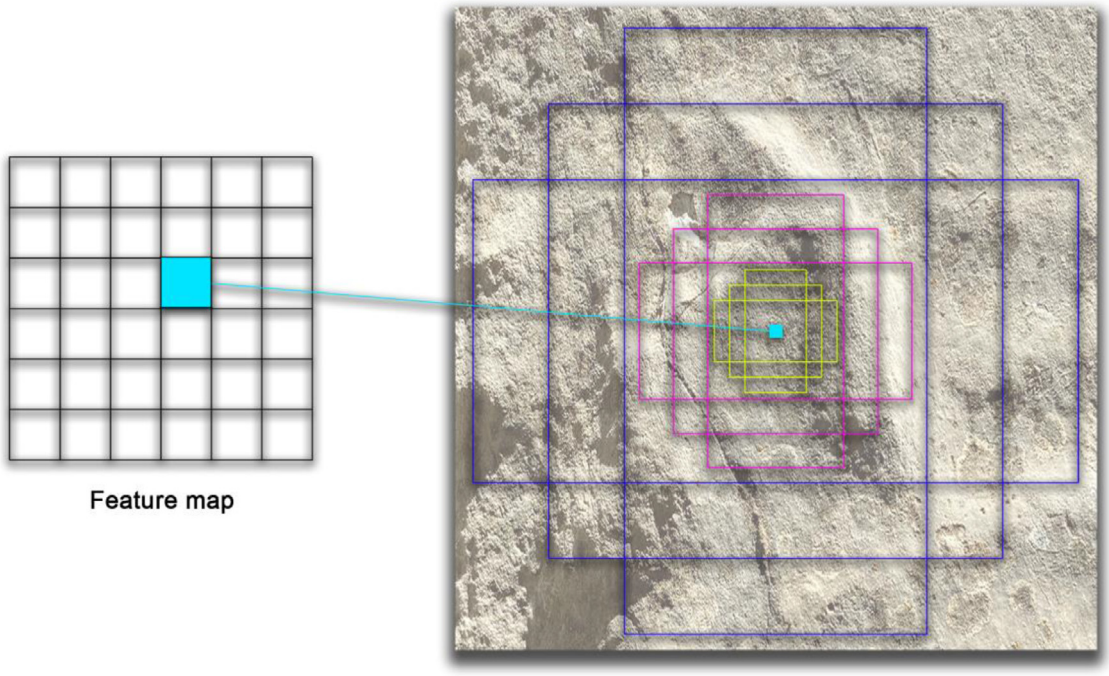


Fig. 5. Representation of the method's approach to nine anchors on the image.

Table 1
Labeling numbers of deterioration and restoration intervention.

Class	Number of labeling
Microcarst	4756
Biological colonization	3028
Deposit	2196
Crack	1064
Contour scaling	893
Missing part	697
Higher plant	541
Impact damage	328
Restoration intervention	197
Total	11,504

ages so that the algorithm can learn the objects. At the validation stage, the algorithm applied to the training data set is improved. The test process is the stage at which the model developed by the algorithm is determined by using different images to determine how close it is to reality. In this study, images were taken from distances varying between 10 cm and 3 m to increase the recognition success of models on different scales (mm-m) in deterioration types and restoration interventions. Moreover, images were gathered to increase the accuracy of the models in different angles, light, and shadow situations. After compiling the data, all images were converted to 800 × 600 pixels to reduce the training time of the algorithm. For the training stage of the model, the coordinates and descriptions of the objects in the images were labeled using the open-source LabelMe software. Of the images labeled for the deposit model via LabelMe, 576 were used for the training data set, and 144 were used for the test and validation data sets. For the other model, the number of images labeled for the training, test, and validation data sets is 1392 and 348, respectively. The labeling numbers of all data sets are presented in Table 1. When Table 1 is examined, it is observed that the difference in the labeling numbers of the classes is directly related to the types of deterioration developing on the rock surface. While microkarst due to chemical deterioration observed in the carbonate rocks where the monuments are located has the highest number of labeling, restoration interventions in the reliefs constitute the lowest number due to its limited number. Afterward, all the labeled images were converted to the .json format and made ready for the training of the models.

The Mask R-CNN method was used in the TensorFlow framework at the training stage of the models. The Faster R-CNN method was used in the head architecture of Mask R-CNN, and ResNet-101 was preferred for the backbone. The batch size, learning rate, and momentum coefficient values were determined to be 16, 0.003, and 0.9, respectively, in the training of the models. The number of iterations indicating the number of weight updates was selected as 26,000 for both models in this study. At the test stage, to determine the capability of the models at different distances, the orthophotography image of 3 reliefs (so-called sword god in chamber B, a row of the female gods in chamber A, and 12 running gods) created at a far distance and the close-range image of one relief (a Luwian Hieroglyph being part of the Šarruma/Tuthaliya IV relief in chamber B) were used. Furthermore, an orthophotography image of the Eflatunpınar water monument was used to evaluate the test success of the models in different regions and stones. The orthophotographs of the monuments were obtained using the PhotoScan program of the sequential images of the reliefs. To accelerate the training and test stages of the proposed model, a computer with a Graphics Processing Unit (GPU) processor and NVidia GeForce GTX 1060 GPU graphic card was used.

5. Experimental results

The total loss graphs showing how far the estimates of the models developed in the study at the end of the training phase are from the ground truth are presented in Fig. 6. The loss value approaching 0 increases the model's estimation reliability, and it is observed to approach 0.1 for 26,000 iterations in the model developed for the deposit (Fig. 6a). In model 2, this value approaches 0.23 (Fig. 6b). The training success of the models was determined by average precision (AP) and mean average precision (mAP), which are among the most common evaluation indicators. AP is the score obtained during the training phase of the model by

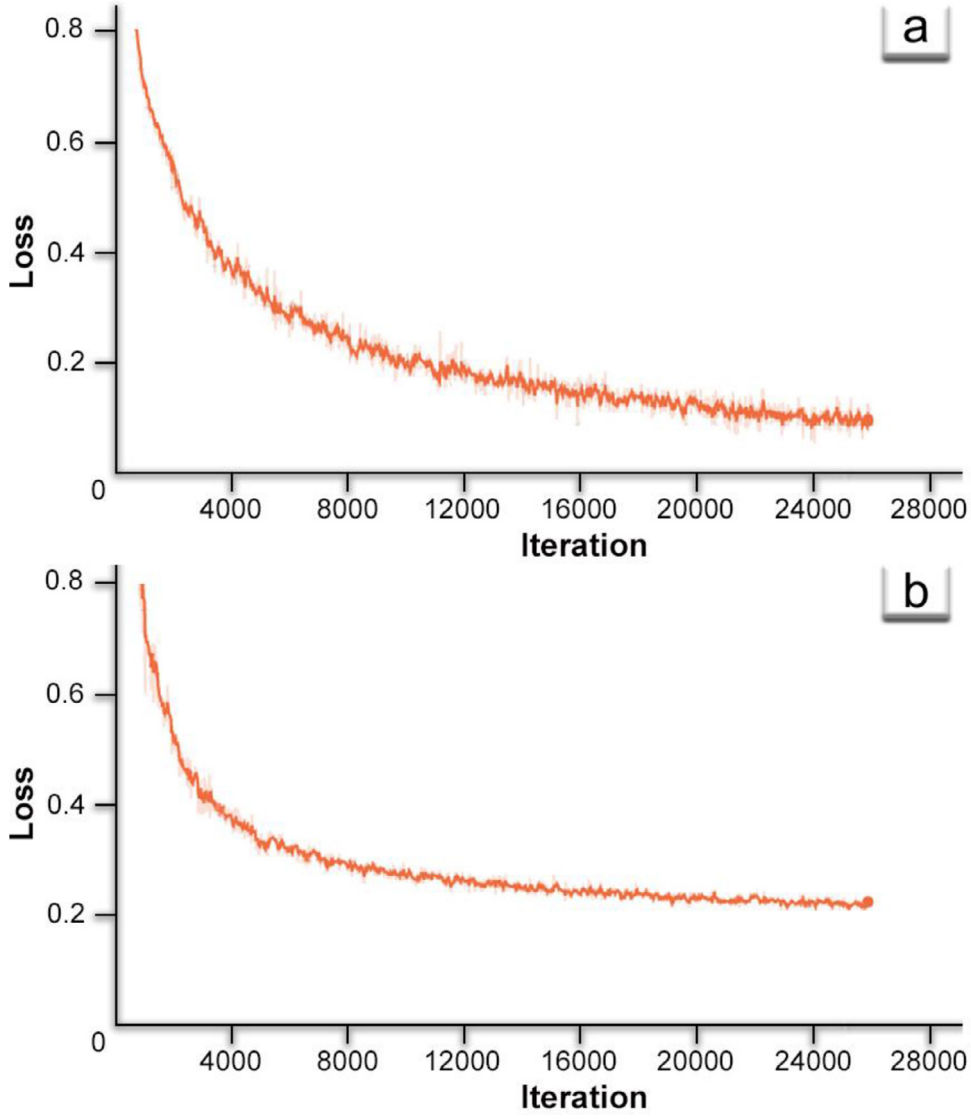


Fig. 6. Total loss curves for models; (a) model 1, (b) model 2.

comparing the ground truth with the detected box of the objects in the categories. This score is defined as positive when there is deterioration and restoration application for the developed model and negative when it is not. If the surface is positive and detection is positive, it is detected as true positive (TP). If the surface is negative and detection is negative, it is detected as true negative (TN). If the surface is positive and detection is negative, it is detected as false negative (FN). If the surface is negative and detection is positive, it is detected as false positive (FP). Using the data obtained from these cases, precision and recall values are found from Eqs. (1) and 2, respectively.

$$\text{precision} = \frac{TP}{TP + TN} \tag{1}$$

$$\text{recall} = \frac{TP}{TP + FN} \tag{2}$$

The AP value of a category is calculated from the area under the precision/recall curve. On the other hand, mAP is calculated as the average of the AP values belonging to the categories in the model. If these two values are close to 1, it indicates that the detection of the model is successful. The AP values showing the accuracy of the deterioration and restoration intervention classes in iterations are given in Table 2. While the highest AP values among these classes (100%) were determined in deposit, microkarst, biological colonization, crack, restoration intervention, and higher plant, the lowest value was revealed in impact damage (AP: 89.624%). The mAP values expressing the average of the AP values of the classes in the models and the training times are given in Table 3. When Table 3 was examined, the mAP values were 100% for model 1, and they were determined to be 97.906% in the other model.

The high success rates obtained from both models were examined on orthophotographs created from three long distances and the Hittite reliefs of one close-range image at the test stage. Among these reliefs, the orthophotograph image of the monument's so-called sword god in chamber B and the results of the model's determination are presented in Fig. 7. When Fig. 7a was examined, it was observed that the deterioration process accelerated in this part due to the high water content in the infiltration zone located in the upper part of the monument and the missing part formation developed as a result of this process. After these losses, deposit formations developed on the relief's surface due to the combination of infiltration water and atmospheric pollutants. Furthermore, the organic substances accumulated in the rock discontinu-

Table 2
Average precision (AP) values obtained from the deterioration categories as a result of the training phase of the models.

Iteration	AP (%) for deposit	AP (%) for microkarst	AP (%) for biological colonization	AP (%) for crack	AP (%) for restoration intervention	AP (%) for higher plant	AP (%) for contour scaling	AP (%) for missing part	AP (%) for impact damage
26,000	100	100	100	100	100	100	97.232	92.725	89.624

Table 3
Mean average precision (mAP) values obtained from the models as a result of the training phase.

Iteration	mAP(%) for model 1	mAP (%) for model 2	Time for Model 1(hour)	Time for Model 2 (hour)
26,000	100	97.906	3:07:49	3:52:03

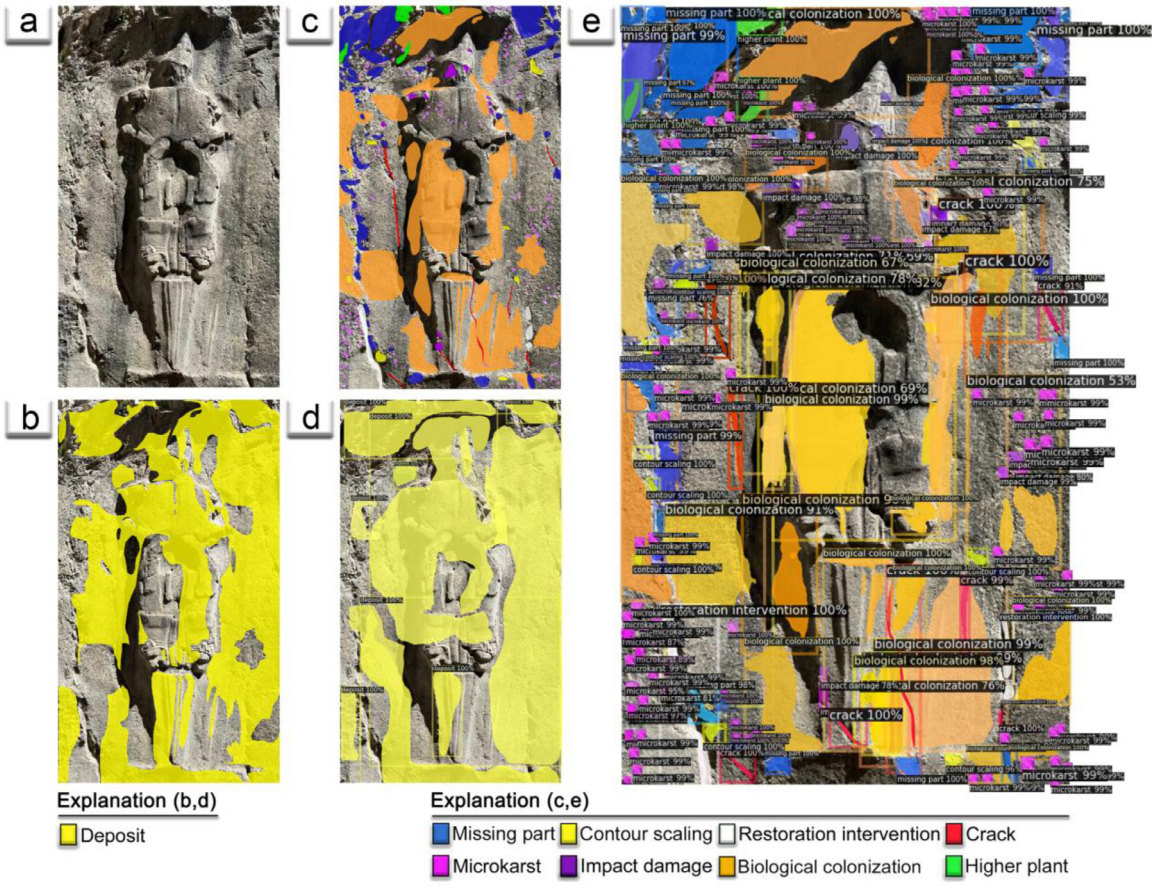


Fig. 7. Detection and mapping results of the relief of the so-called sword god in chamber B; (a) orthophographic image, (b) vision-based map of deposit deterioration type, (c) vision-based maps of restoration intervention and other deterioration type, (d) test result of model 1, (e) test result of model 2.

ities caused the formation of higher plants due to the active water on the surface, while biological colonization developed widely throughout the monument. The maps prepared by the visual analysis of the types of deterioration in the monument are given in Fig. 7b and c. The images of the test results obtained from model 1 and model 2 are presented in Fig. 7d and e, respectively. From these images, when Fig. 7d is examined, it is observed that the deposit anomaly covering a large part of the relief surface could be determined with 100% accuracy. The anomalies with the lowest and highest accuracy rates obtained from the estimates of the types of deterioration in the monument were determined to be biological colonization (53–100%), impact damage (78–99%), microkarst (87–100%), crack (91–100%), contour scaling (96–100%), missing part (97–100%), higher plant (100%), and restoration intervention (100%), respectively (Fig. 7e). It is thought that biological colonization with the lowest accuracy rate in model 2 was due to its similarity to the deposit anomaly covering the relief surface to a significant extent and the scattered distribution of the anomaly on the deposit surfaces. Despite a considerable similarity when comparing the maps obtained from the visual and algorithm estimates of the deposit deterioration type, the contours of the DL results were determined to be smoother (Fig. 7b and d). When other maps were evaluated, all deterioration types in the visual analysis were compatible with the DL algorithm.

When the monument's orthophotograph of a row of the female gods in chamber A is examined, it is observed that the most prominent deterioration factor is the crack limiting the relief and passing through its middle (Fig. 8a). While higher plant developments were observed on the crack passing through the middle of the monument, restoration was applied to the lower part of the crack. The natural rock protrusion in the upper part of the monument prevented water from infiltrating the relief surface. This process limited deposit formation on the relief surface. While limited biological colonization and higher plant formation are observed on

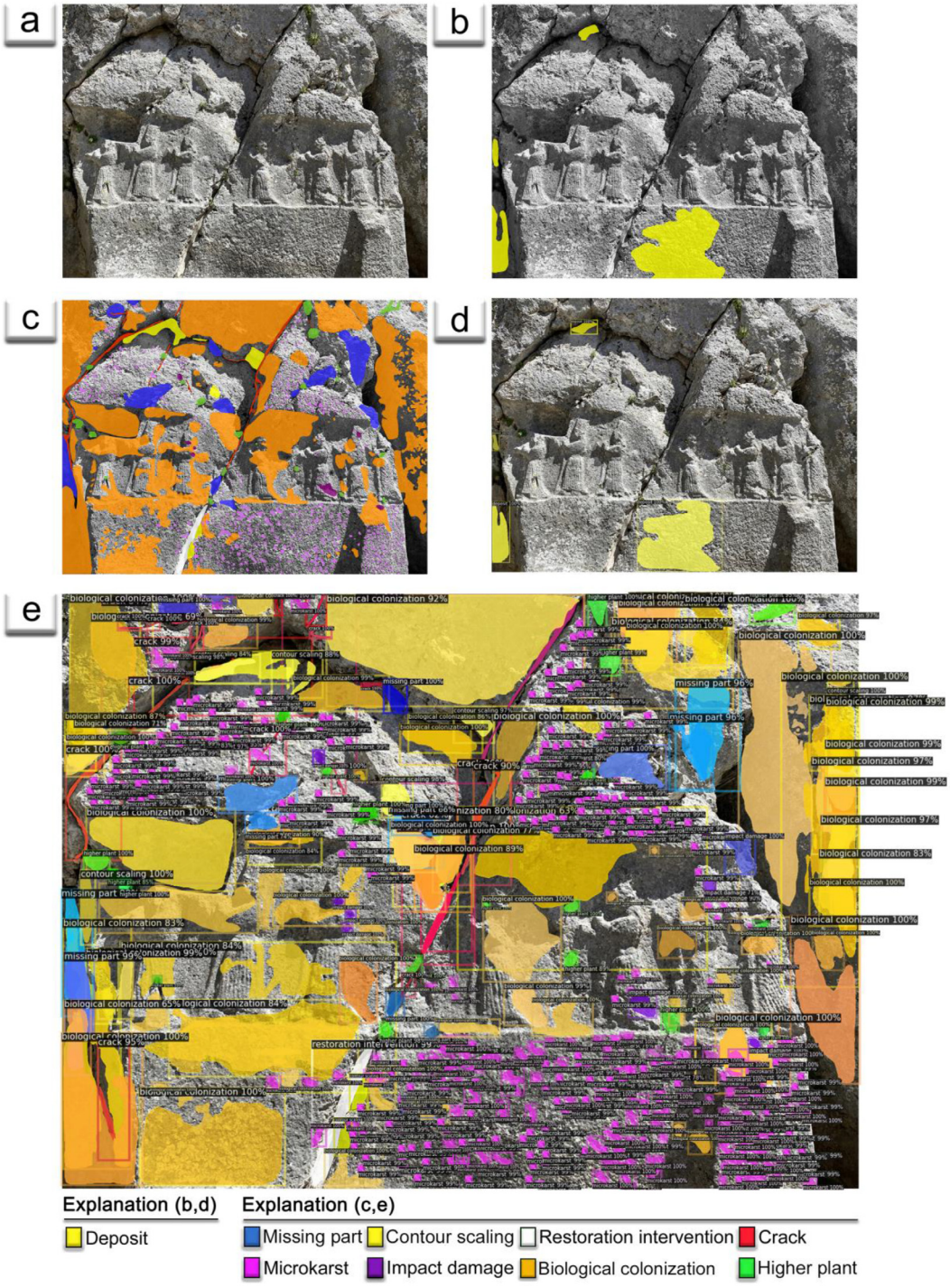


Fig. 8. Detection and mapping results of the reliefs of a row of the female gods in chamber A; (a) orthophographic image, (b) vision-based map of deposit deterioration type, (c) vision-based maps of restoration intervention and other deterioration type, (d) test result of model 1, (e) test result of model 2.

the relief surface, there are microkarst concentrations in the other parts. The visual analysis of all these types of deterioration in the monument is presented in Fig. 8b and c, while the estimates of the algorithms are given in Fig. 8d and e. When the test results of model 1 were evaluated, the limited deposit anomaly in the monument was determined with 100% accuracy (Fig. 8d). When the results of the deterioration and restoration interventions of model 2 in which this monument was tested were examined, the lowest accuracy rate was determined in biological colonization (65–100%). The accuracy rates for other deterioration types are 92–100% in missing part, 84–98% in contour scaling, 76–100% in crack, 76–100% in microkarst, and 71–100% in impact damage. It is 85–100% in the plant and 99% in the restoration intervention (Fig. 8e). When visual analysis and computer vision results were evaluated together, it was observed that the types of deterioration completely overlapped. Moreover, natural outcroppings in the rock at the top of the monument were not defined as missing part by model 2. However, significant attention should be paid to such cases since natural outcropping can be defined incorrectly in the insufficient data set.

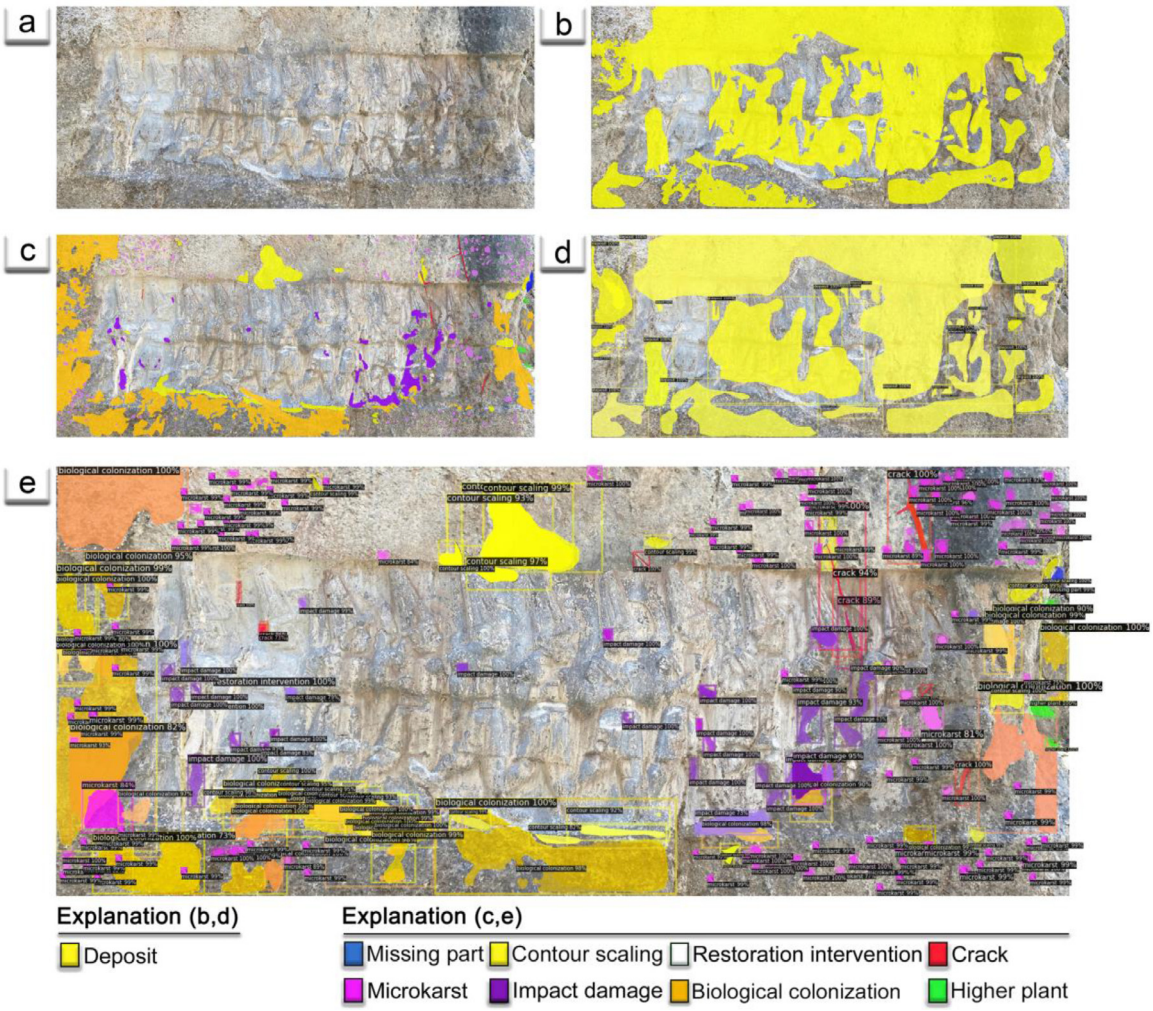


Fig. 9. Detection and mapping results of the relief of the 12 running gods; (a) orthophographic image, (b) vision-based map of deposit deterioration type, (c) vision-based maps of restoration intervention and other deterioration type, (d) test result of model 1, (e) test result of model 2.

Impact damage and deposit type deteriorations are common on the relief surface of the 12 running gods in chamber B, which is the last one of the orthophotography images created for testing the models (Fig. 9a). Impact damage, one of these types of deterioration, is thought to result from the lack of necessary protection measures in the Hattusa site until the recent period. Other deterioration anomalies were concentrated around the relief. Maps based on the visual inspection of deterioration anomalies observed on the relief are given in Fig. 9b and c. The test results of the models are presented in Fig. 9d and e. While the accuracy rates of the deposit model in Fig. 9d are 100%, the accuracy rates in the classes of the other model vary between 73 and 100%. The accuracy rates for crack, impact damage, biological colonization, microkarst, contour scaling, missing part, restoration intervention, and higher plant deterioration types in model 2 were estimated to be 73–100%, 73–100%, 73–100%, 81–100%, 82–100%, 99%, 100%, and 100%, respectively (Fig. 9e). When the maps (visual inspection and DL) of this monument were evaluated together, the positions of all deteriorations were determined by taking them correctly. However, in DL estimates, the contours of deposit and biological colonization types showed smoother transitions. Furthermore, the proposed algorithm evaluated the region where microkarst types were concentrated in the lower left corner of the relief as a whole.

The image of a Luwian Hieroglyph being part of the Šarruma/Tuthaliya IV relief in chamber B used to test the success of the models at a close distance is given in Fig. 10a. Upon examining this image, while deposit and microkarst are quite common throughout the relief, there are also missing part, contour scaling, impact damage, biological colonization, and restoration intervention classes. The visual inspection and DL algorithm maps in this relief are given in Fig. 10b-e. As in the results of model 1 in other monuments, the deposit anomaly was determined with 100% accuracy in this relief (Fig. 10d). When the test results of model 2 for this monument were examined, the microkarst anomaly with different sizes on the relief surface was determined at an accuracy rate varying between 75% and 100% (Fig. 10e). Accuracy rates in other types of deterioration were determined as 68–100% in impact damage, 68–100% in biological colonization, 78–100% in missing part, 82–100% in contour scaling, and 99–100% in restoration intervention (Fig. 10e). When visual investigations were compared with DL algorithms, the suggested models accurately predicted all types of deterioration, except for biological colonization and deposit. In the models, biological colonization in the upper part of the monument was detected as both deposit and biological colonization. Anomalies in this region of the relief caused detection problems in the algorithm due to the intense shadow. In this sit-

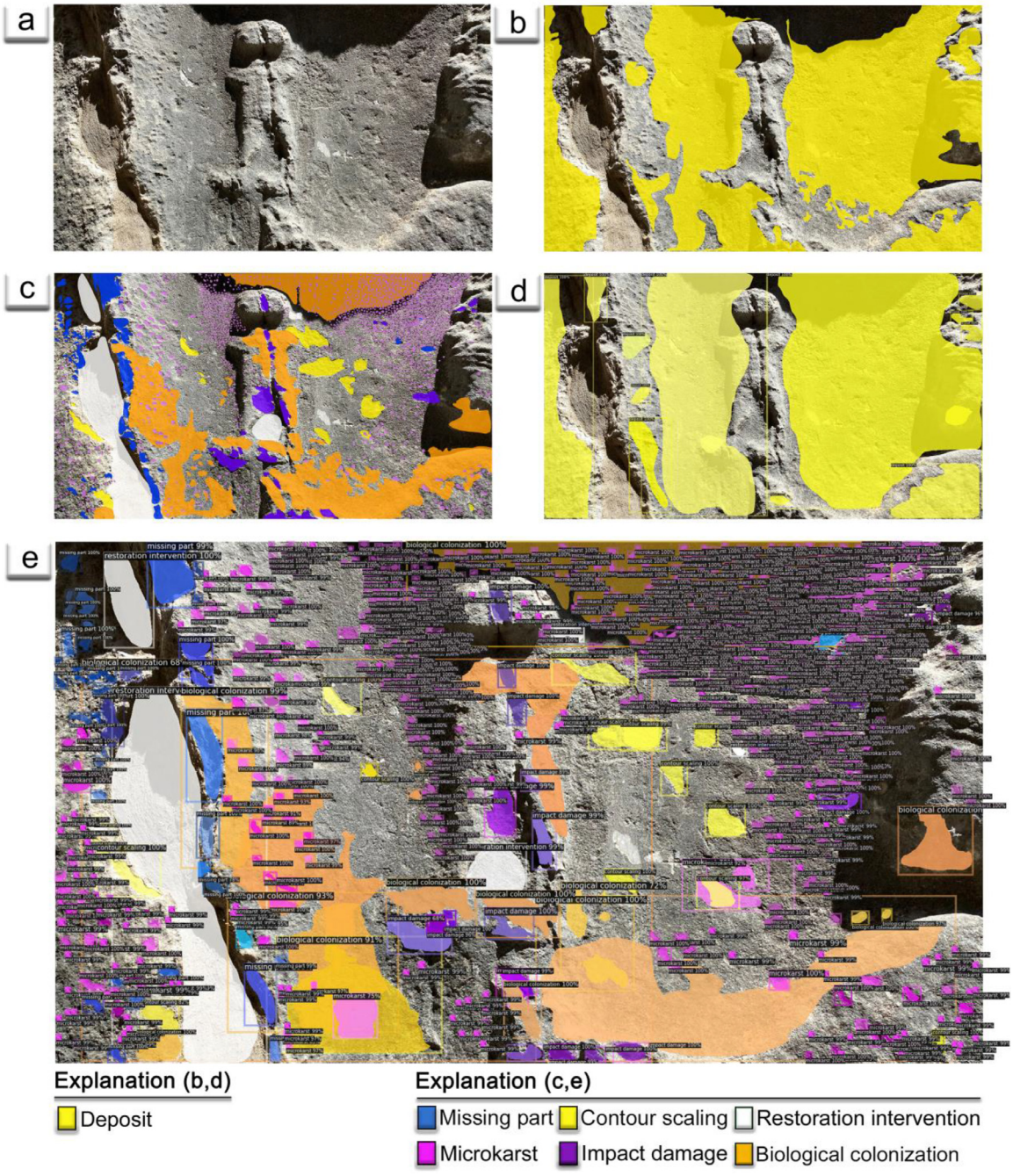


Fig. 10. Detection and mapping results of a Luwian Hieroglyph being part of the Šarruma/Tuthaliya IV. Relief in chamber B; (a) original image, (b) vision-based map of deposit deterioration type, (c) vision-based maps of restoration intervention and other deterioration type, (d) test result of model 1, (e) test result of model 2.

uation, when both anomalies are difficult to distinguish even with human eyes, the algorithm developed made two different suggestions for the same location. These suggestions can be considered positive since they will enable conservation experts to focus on this area.

To evaluate the universal usability of the developed model, the image of the Eflatunpınar water monument built in a different climatic environment and lithology was used (Fig. 11a). The common types of deterioration in this monument are biological colonization caused by rainfall in the infiltration zone of the structure and higher plant formations in the capillary region. There are impact damage and contour scaling anomalies between these two regions. Additionally, rounding type deterioration, not found in the Yazılıkaya reliefs, is evident in the figures on the blocks. The mapping of all these deteriorations observed in the monument with visual analysis is given in Fig. 11b. Since there is no deposit deterioration type in this structure, it was tested only with the model 2 algorithm (Fig. 11c). When Fig. 11c was examined, biological colonization, the most widespread type of deterioration in the monument, was estimated between 58 and 100%, while the higher plant formation was determined with accuracy rates between 56 and 100%. The accuracy rates of crack, missing part, impact damage, and contour scaling, among other anomalies, were determined as 100%, 100%, 93–100%, and 75–100%, respectively. When Fig. 11b prepared according to the visual analysis and Fig. 11c obtained from model 2 are compared, the biggest difference is the rounding

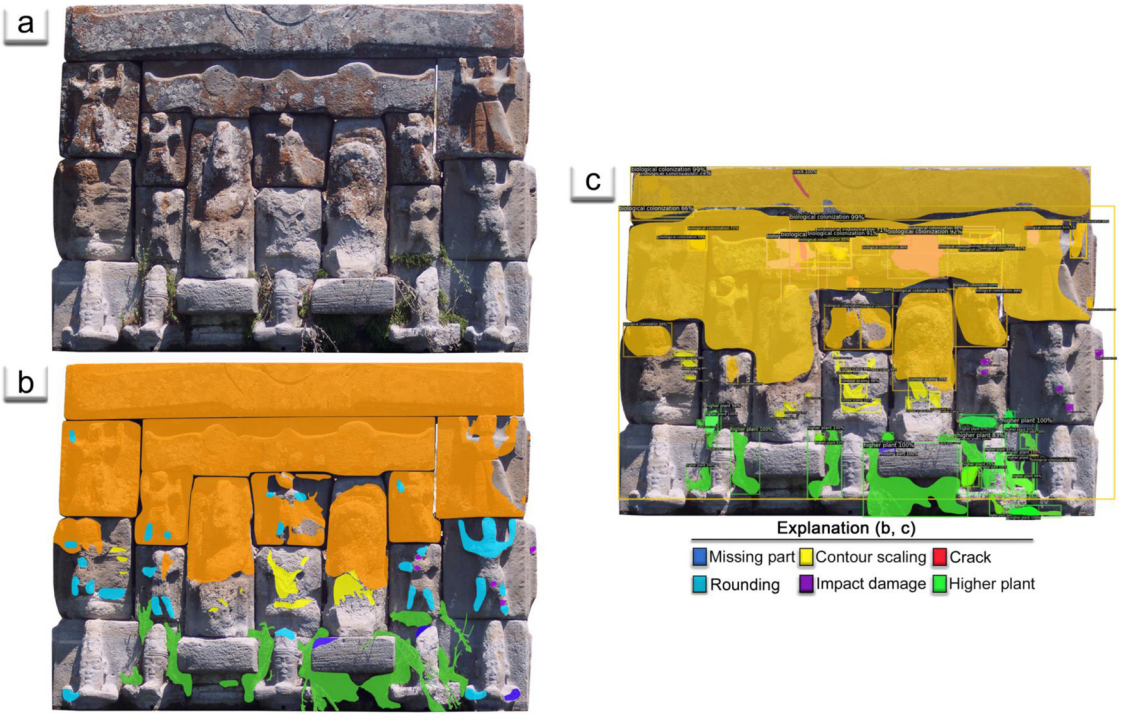


Fig. 11. Detection and mapping results of Eflatunpınar water monument; (a) original image, (b) vision-based map of deterioration, (c) test result of model 2.

anomaly not included in the training set of the algorithm. Furthermore, a few minor errors were found in the model. The first one of these errors is that the crack type not observed in the structure was detected in the model. This is because the figure on the building block was covered with biological colonization over time. The other problem of detection is that only one of the missing part types found in three different parts of the monument could be detected. This problem arises because the other two missing part types become rounded over time and appear similar to the original figure.

6. Conclusion

The periodic control of deterioration in archeological heritage sites is extremely important for preserving artifacts and transferring them to future generations. However, due to the size of archeological sites and the necessary vision-based controls, the duration of examination increases, which may cause the occurrence of possible human-induced errors. In this study, two different computer vision models were developed using the deep learning algorithm to prevent possible errors and automatically create the updated maps of deterioration and restoration interventions. To this end, the images of 8 deterioration types in Hattusa Yazılıkaya monuments and restoration practices were trained using the Mask R-CNN-based algorithm models and tested on four unique rock reliefs. Furthermore, an image of the Eflatunpınar water monument was used to determine the model's universal usability.

In the study, 26,000 iteration steps were used to train both models. In this iteration step, the mAP value for model 1 was 100%, while this value was 97.906% for the other model. The AP values of the classes of the models varied between 89.624% and 100%.

The ability to determine very different anomalies with a different size (mm-m), surface (planar, undulating, and stepped), shape (linear like crack, planar like contour scaling, concave formation like microkarst), and color at a very high success rate in the proposed models was evaluated as a success criterion of the program.

It was compared with the maps obtained from the visual analysis to evaluate the success of models in determining types of deterioration with restoration interventions. Both long-distance (orthophotographs) and close-range test results of the proposed models have considerable similarities with the visual analysis. This indicates that DL models can perform rapid and automatic deterioration mapping in archeological sites. The developed algorithm can be considered a helpful method for conservation experts.

The algorithms developed within the scope of the study are the first model with the feature of Mask R-CNN-based deterioration identification and mapping in stone heritage sites. The model was tested in monuments in two different regions belonging to different environmental conditions, stone type and construction technique, and its universal usability was proven. Owing to the development of this approach, it is obvious that it can easily be transformed into a practice that can be continuously monitored with mobile or computer applications which can be used effectively in all stone heritage sites and in which change can be followed quickly. Moreover, the time of experts in the field can be optimized by the autonomous UAVs to be integrated into this method, which is extremely fast in deterioration detection and mapping, in future studies.

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